LESSON 2.10

Choose Appropriate Measures

Start here · Words, worked example, and sentence starters

Name _____ **Date** _____ **Period** _____

LEARNING TARGET I can choose the best measure of center for a data set based on its shape.

WHY THIS MATTERS The students want to describe the data they collected about the number of years of teaching experience of the teachers in their school. Which measure of center and measure of variation should they use to summarize the data set?

START HERE

Words for this lesson

Read each word before you begin. Say it out loud.

WORD	WHAT IT MEANS	EXAMPLE
Mean *Spanish: Media*	The average. Add all the numbers, then divide by how many there are.	Mean of 10, 20, 30 = (10+20+30) ÷ 3 = 20
Median *Spanish: Mediana*	The middle number when you put them in order.	Data: 5, 8, 12, 15, 20 → median is 12 (the 3rd of 5 values)
Outlier *Spanish: Valor atípico*	A number that is much bigger or smaller than the rest.	2, 3, 4, 32 → 32
Skewed *Spanish: Sesgado*	When most data sits on one side with a tail on the other.	A long tail to the right
Data distribution *Spanish: Distribución de datos*	How the data looks: where it sits and how spread out it is.	Symmetric = even on both sides. Skewed = bunched on one side with a tail
Variability *Spanish: Variabilidad*	How spread out the numbers are.	88, 90, 89, 91 (low variability) vs. 50, 70, 95, 100 (high variability)

HOW IT WORKS

Worked example

These numbers are not on your problems. The steps are. Follow them with your own numbers.

Five friends saved $6, $7, $8, $9 and $50. Which center describes a typical friend?

1. 50 is an outlier.
2. Median = 8. Mean = 80 ÷ 5 = 16, more than four of the five values.
3. The median is closer to most of the values.

Answer: The median, $8

NOW YOU TRY

Same steps, your turn

The steps are the same as the worked example. The numbers are yours. Do the work.

Data set A: 4, 5, 6, 7, 8. Are there any outliers? If not, which measure should we use, mean or median?

1. _____
2. _____
3. _____

Answer: _____

SAY IT AND WRITE IT

Sentence starters

Finish each sentence out loud with a partner. Then use them in your writing.

- The outlier in this set is _____ because _____ .
- The median is _____ and the mean is _____ .
- The mean is pulled _____ because _____ .
- I would report the _____ because _____ .

Word bank outlier | skewed | cluster | gap | median | mean | typical | center | pulled | symmetric | describe | data set

WATCH OUT!

A common mistake

A common mistake in Appropriate Measures is defaulting to the mean just because "it uses all the data," without first checking whether an outlier or skewed shape is present. For example, given a player's points per game 14, 16, 15, 17, 16, 58, a student calculates the mean (136 ÷ 6 ≈ 22.7) and reports that as the typical game

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Choose Appropriate Measures

Level 0 · Most support · every hint printed

MASTERY CHECK
- ☐ Exceeds
- ☐ Meets target
- ☐ Needs practice

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PART 1 Understand the idea

models and meaning

1 **Write the letter of the matching item on each line.**

Match each data description with the best measure of center.

_____	Test scores: 88, 90, 85, 92, 87 (no outliers)	A	Mean
_____	Salaries: $30K, $32K, $35K, $31K, $250K	B	Median
_____	Race times: 12.1s, 12.3s, 12.0s, 12.2s, 12.1s	C	Mean
_____	Goals scored: 1, 2, 1, 0, 2, 1, 15	D	Median

STEPS TO TRY

1. Scan each data list for a value that's wildly different from its neighbors — like one huge salary or one giant goal count.
2. Rule: outlier present → the middle value represents 'typical' better; no outlier → the average works fine. Decide outlier-or-not for each row.
3. Start with the salaries: $250K towers over the four values near $30K — a clear outlier. Match that row, then test the other three rows.

PART 2 Apply it

real-world problems

2 **Circle the letter of the best answer. Show how you know.**

A gymnast's scores are: 8.5, 8.8, 8.7, 8.6, 8.9. There are no outliers. Which measure best represents a typical score?

(A) Median

(B) Neither — the data is too small

(C) Both are equally bad

(D) Mean

STEPS TO TRY

1. Look at the scores: 8.5 up to 8.9 — all five are bunched tightly together with no unusual value.
2. Rule of thumb: with no outliers and balanced data, one measure uses every value fairly. With outliers, the middle value is safer. Which case is this?
3. First step: check for an outlier — is any score far from the others? All five sit within 0.4 points, so apply the no-outlier rule.

SHOW YOUR WORK

3 **Circle the letter of the best answer. Show how you know.**

A runner's mile times are: 7:10, 7:15, 7:12, 7:20, 12:00. The 12:00 was due to a cramp. Which measure best represents a typical mile?

(A) Mean — it uses all the data

(B) Median — the outlier 12:00 pulls the mean too high

(C) Mode — it shows the most common time

(D) Neither measure works

STEPS TO TRY

1. Four times are near 7 minutes; one is 12:00 from a cramp. That 12:00 is an outlier, not a typical mile.
2. Ask: which measure gets dragged toward an extreme value, and which one just takes the middle of the ordered list?
3. First step: order the times and notice where 12:00 sits — far beyond the rest. Then choose the measure that isn't pulled by that extreme time.

SHOW YOUR WORK

4 **Circle the letter of the best answer. Show how you know.**

Data set: 5, 6, 7, 7, 8, 50. The mean is about 13.8 and the median is 7. Which better represents the typical value?

(A) Mean (13.8) — it includes all values

(B) Both are equally good

(C) Median (7) — the outlier 50 inflates the mean

(D) Neither works for this data

STEPS TO TRY

1. Five of the six values are between 5 and 8 — then there's 50. Notice where the mean (13.8) lands compared to those five.
2. Test for 'typical': count how many actual data values sit near 13.8, then how many sit near 7. A good measure stays close to most of the data.
3. First step: compare 13.8 to the list — it's bigger than 5, 6, 7, 7, AND 8. Ask what pulled it up there, and whether the median has that problem.

SHOW YOUR WORK

EXPLAIN YOUR THINKING

What pattern do you notice about when median is the better choice?

SENTENCE STARTER The median is better when the data has ___ because the mean gets pulled toward ___. The mean is better when the data is ___ because ___.

3-Point Exit Checklist: ☐ 1. Units & Labels Verified ☐ 2. Mathematical Reasoning Stated ☐ 3. Calculations Double-Checked

How did it go? ☐ 4 · I can teach it ☐ 3 · I've got it ☐ 2 · I need a hint ☐ 1 · I need help